

Use of Edge Machine Learning for a Non-Invasive Beehive Monitoring System

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Abstract—Climate change poses significant threats to honeybee populations, making continuous monitoring of hive activity crucial for beekeeping management and ecological research. This paper presents an innovative non-invasive bee monitoring system that combines machine learning and computer vision techniques for automated bee counting at hive entrances. The system utilizes a YOLOv8n object detection model, deployed on a Raspberry Pi Zero 2W embedded platform, enabling local processing without requiring cloud transmission. Performance evaluation demonstrated a Mean Average Precision (mAP50) of 0.986, with a precision of 0.91, a recall of 0.96, and an F1-score of 0.94 in bee detection tasks. Gold-standard validation using Pearson correlation analysis showed strong agreement ($r = 0.93$) between automated and manual counts. Bland-Altman analysis revealed a small mean bias of 3.4 bees with 95% limits of agreement within ± 1.96 SD. The automated approach eliminates manual observation requirements, reduces labor costs, and colony disturbance while maintaining high measurement accuracy, offering significant advantages for precision beekeeping and ecological monitoring initiatives.

Keywords — Honeybee, Bee behavior analysis, Insect detection, Machine learning, Convolutional neural network, Computer vision, Picture analysis, YOLOv8, Edge computing, Non-invasive monitoring.

I. INTRODUCTION

The honeybee (*Apis mellifera* L.) holds substantial economic importance in both agricultural and natural ecosystems due to its crucial role as a pollinator [1]. The species distribution and biological traits are strongly influenced by climatic variables, particularly seasonal fluctuations in rainfall and temperature [2]. Currently, profound alterations in global climate patterns are being observed, with projections indicating that these changes are expected to persist in the coming decades. Such shifts are expected to affect the diversity of plants and animals, including bees, as demonstrated by numerous simulation studies conducted under different climatic scenarios [3–5].

In response to these challenges, this paper describes an innovative bee monitoring system that combines machine learning and computer vision techniques to estimate local bee numbers at hive entrances. By performing all processing locally without requiring cloud transmission, the system achieves significant reductions in communication latency, bandwidth consumption, and operational expenses. The first section outlines the fundamental concepts of machine learning, including its prerequisites, capabilities, and limitations. The second section presents an overview of dataset creation, accompanied by an introduction to the system and a description of model training. The final section discusses potential applications and includes a brief demonstration of the bee-counting system. Among the YOLOv8 [6] variants evaluated, YOLOv8n was selected as it delivers sufficient detection accuracy while maintaining optimal computational efficiency within our hardware limitations. Subsequently, the gold standard methods are applied to compare and critically analyze the results. We anticipate that this work will facilitate the integration of such approaches into practical applications while also highlighting their potential to researchers in non-technical disciplines.

II. MACHINE LEARNING

Machine learning is a subfield of artificial intelligence (AI). This term encompasses a wide array of methods, models, and algorithms aimed at emulating human cognitive processes across diverse domains, including computer science (e.g., computer vision, classification, and recognition tasks), physics (e.g., modeling physical processes), pharmacology, and biology (e.g., drug and molecule discovery) [7]. The concept of artificial intelligence has been defined in multiple ways and is often misinterpreted. For this reason, the present work focuses specifically on machine learning.

Machine learning can be defined as a branch of AI that seeks to replicate the human capacity to learn autonomously

from data by identifying patterns, structures, or clusters within it [8]. Approaches in this field may rely on statistical modeling, clustering techniques, mathematical transformations, or deep learning architectures. In nearly all cases, however, the performance of these methods depends critically on the quality and structure of the data (datasets, as described below) [9]. Furthermore, successful learning in one context does not guarantee satisfactory results when applied to different datasets.

A common classification of machine learning methods is based on the learning paradigm: supervised (which was applied in our model) or unsupervised. This distinction depends on whether the input data are labeled before training. In supervised learning, labeled data require greater effort during the preparation stage but typically yield more reliable and accurate results compared to unsupervised methods. However, in situations where data labeling is infeasible, unsupervised learning must be employed [10].

III. METHODOLOGY

A. Dataset

To develop a model capable of detecting and counting bees at hive entrances, a dataset of annotated images is required in which each bee is labeled with its precise location. The primary motivation for compiling such a dataset was to enable non-invasive monitoring of multiple hives without disturbing natural colony activity.

The dataset comprises 4,574 individual frames (images) collected from two different beehives. These images were split into training (87%), validation (9%), and testing (4%) datasets. All frames were captured at a resolution of 1920×1080 pixels and subsequently compressed to 640×640 pixels due to YOLO's resolution requirements. We uploaded the images to Roboflow [11] a comprehensive computer vision platform that streamlines dataset management and annotation workflows, to facilitate the labeling process (Figure 1).

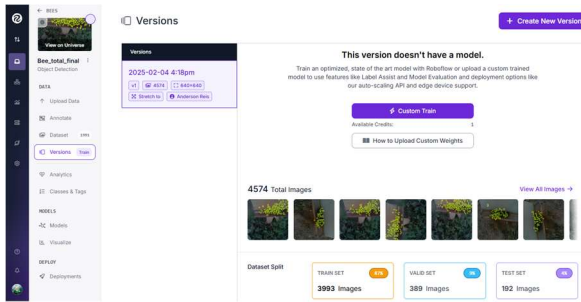


Fig. 1. Roboflow environment.

All annotations across the proposed datasets were stored in YOLO format. The labeling process was semi-automated: a Roboflow 3.0 Object Detection (Fast) model was first trained on a manually annotated subset for bee detection. Afterward, the generated labels were visually inspected and corrected as needed. The frames were captured using an OV2640 camera within the XIAO ESP32S3 Sense board. With a horizontal field of view (FoV) of 53.5°, a camera positioned at the top of a standard hive (46 cm in height) can capture the entire entrance (approximately 47 cm in width).

Figure 2 illustrates the dataset, which includes samples from two distinct beehive entrances where individual bees are annotated with rectangular bounding boxes for the bee detection dataset. All visible bees within each frame were annotated regardless of their position—whether located on the hive's front surface, entrance platform, or adjacent vegetation and background regions. In instances where bees partially overlap, their corresponding bounding boxes also overlap accordingly. When only a portion of a bee is visible, either due to its transition through the entrance aperture or partial appearance at the frame boundary, annotations are applied exclusively to the visible regions.



Fig. 2. Images of two different beehive entrances, comparing labelling in the bee detection dataset.

B. Hardware

Figure 3 illustrates the system architecture, which comprises several key components. The core processing unit consists of a Raspberry Pi Zero 2W featuring a 1GHz quad-core 64-bit ARM Cortex-A53 CPU and 512MB of SDRAM. The device incorporates 2.4GHz 802.11 b/g/n wireless LAN, Bluetooth 4.2, and Bluetooth Low Energy (BLE) capabilities with an integrated antenna for enhanced connectivity. Input/output interfaces include a Mini HDMI® port for visual outputs and a micro-USB On-The-Go (OTG) port for data connections. Storage is managed via a microSD card slot, while imaging capabilities are provided through a CSI-2 camera connector. The board features a 40-pin header for system expansions and sensor connections. Power is supplied through a micro-USB port, with composite video and reset pins accessible via solder test points. The compact dimensions of 65mm × 30mm make the device highly suitable for embedded applications.

The OV5647 is a low-voltage, 5-megapixel CMOS image sensor that delivers 2592x1944 pixel resolution. This 1/4-inch optical format sensor features a 1.4µm pixel size and supports multiple output formats, offering flexible frame rates from 30fps at full resolution to 90fps at 480p resolution. The OV5647 camera module features a 54° × 41° field of view with a fixed-focus 3.60 mm lens, providing sharp imaging from 1 meter to infinity. The compact 25×24 mm module operates with a peak current of 300 mA, delivering imaging equivalent to that of a 35 mm full-frame camera lens.

The SHT20 is a fully calibrated digital humidity and temperature sensor IC from Sensirion featuring an I2C interface communication. The sensor delivers typical humidity accuracy of $\pm 3.0\%RH$ across its full relative humidity operating range of 0-100%RH. For temperature measurements, the sensor achieves a typical accuracy of $\pm 0.3^{\circ}C$ and an operating temperature range of $-40^{\circ}C$ to $+125^{\circ}C$.

Timekeeping functionality is implemented using a DS1302 real-time clock (RTC) module. The DS1302 interfaces with the host microprocessor via a simple three-wire serial communication protocol. It maintains comprehensive calendar data, including seconds, minutes, hours, day of the week, date, month, and year information with automatic leap year compensation.

All components were integrated within an IP65-rated enclosure that protects against dust and water ingress, ensuring durability in outdoor environmental conditions. The complete system was mounted on a tripod and positioned directly facing the beehive entrance, as shown in Figure 4, to provide a stable and adjustable platform for data acquisition while minimizing disturbance to the colony.

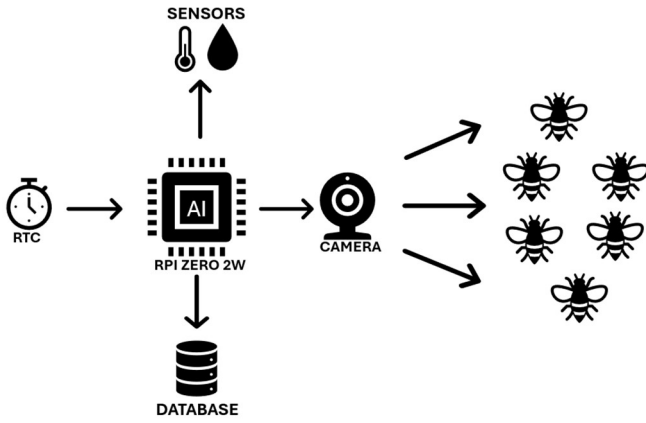


Fig. 3. System architecture.



Fig. 4. System structure.

IV. QUANTITATIVE RESULTS ANALYSIS

After exporting the model in YOLOv8 format, the dataset can be obtained as a compressed archive for local storage and processing. Alternatively, the generated code can be downloaded directly for execution.

The model was trained using an NVIDIA Tesla T4 GPU. This hardware platform is designed for high-performance deep learning workloads and offers an effective balance between computational efficiency and scalability. The Tesla T4 is based on Turing architecture and integrates 2,560 CUDA cores and 320 Tensor Cores. It delivers up to 65 TFLOPS of mixed-precision performance and features 16 GB of GDDR6 memory. The GPU is optimized for both training and inference tasks by supporting large-scale parallelism while maintaining relatively low power consumption (70 W TDP). Due to its efficiency, the Tesla T4 is widely adopted in cloud computing environments, including Google Colab, where it provides accessible and reliable resources for model training and evaluation.

The training process was carefully monitored to ensure convergence of validation metrics. The model was trained for 100 epochs, requiring approximately 2.4 hours to complete. This configuration provided sufficient iterations to achieve convergence while mitigating overfitting risk. Validation metrics progressively stabilized during training, indicating that the model reached an optimal balance between accuracy and generalization.

The results were highly satisfactory. The model demonstrated consistent performance across both training and validation datasets, confirming the robustness and reliability of the final model. Convergence was assessed by the absence of further improvements in the composite validation loss. This loss was defined as the sum of the bounding box localization loss, object classification loss, and distribution focal loss. After training, the model showed a result of (mAP50 of 0.986). Figures 5 and 6 show the confusion matrix and the metrics curves per epoch respectively.

To validate the model, object detection inference was performed using the trained YOLOv8 model. The optimal weights from the training phase were loaded and applied to test images with a confidence threshold of 0.25. The system generated prediction results, including bounding boxes, as shown in Figure 7.

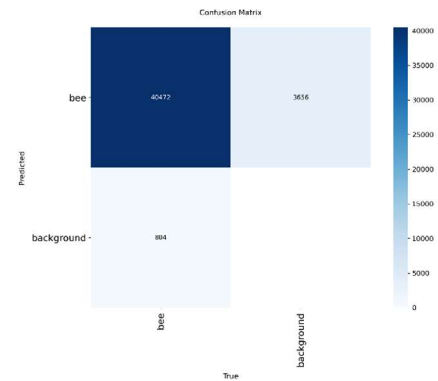


Fig. 5. Confusion matrix.

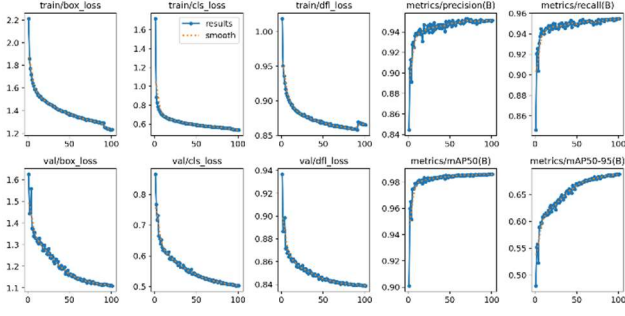


Fig. 6. Training metrics

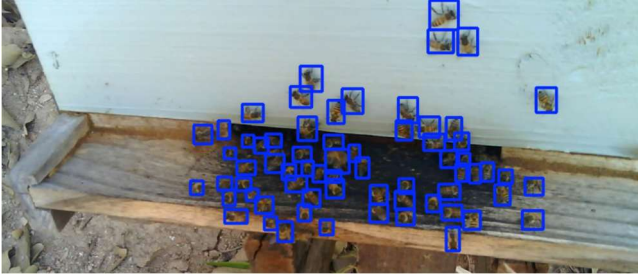


Fig. 7. Model validation

The experimental results presented in Figures 8 and 9 demonstrate that the proposed system successfully detected the majority of bees in the captured images, exhibiting robust performance and high reliability across diverse environmental conditions. Quantitative evaluation revealed a precision of 0.91, a recall of 0.96, and an F1-score of 0.94 in bee detection tasks. However, the analysis identified certain limitations, manifested as false-positive and false-negative detections. False positives predominantly occurred when complex background textures or morphologically similar non-bee objects were erroneously classified as bees. Conversely, false negatives were primarily attributed to challenging scenarios including partial occlusions of individual bees, overlapping bee clusters, and motion-induced blur artifacts that compromised image quality. Despite these limitations, the overall detection accuracy remained sufficiently high to validate the system's practical viability for real-world applications. These results underscore the system's potential as a reliable computational tool for automated monitoring and assessment of beehive activity patterns, offering significant advantages for apicultural research and colony management practices.



Fig. 8. Inference results

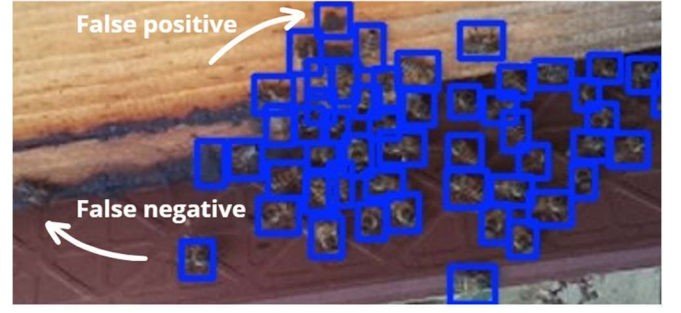


Fig. 9. False positives and false negatives

The system maintained full operational status throughout the entire testing period, with all experimental results successfully stored in the database. Table 1 presents the data file structure, which contains the stored parameters: timestamp, count, inference time, temperature, and humidity. Figure 10 illustrates the temporal variations in count, temperature, and humidity values derived from the recorded data over a 2-hour monitoring period. The average data acquisition cycle, encompassing image capture, inference processing, and database storage, was approximately 2 seconds per record, demonstrating the reliability and computational efficiency of the proposed system architecture.

TABLE I. DATABASE SYSTEM RESULTS

Timestamp	Count	Inference time	Temperature	Humidity
2025-08-23_15-29-44	24	841	26.50°C	34,02
2025-08-23_15-29-46	26	852	26.50°C	34,02
2025-08-23_15-29-49	25	847	26.50°C	34,02
2025-08-23_15-29-51	26	841	26.50°C	34,02
2025-08-23_15-29-53	26	841	26.50°C	34,02
2025-08-23_15-29-55	33	845	26.50°C	34,08
2025-08-23_15-29-57	32	846	26.50°C	34,15

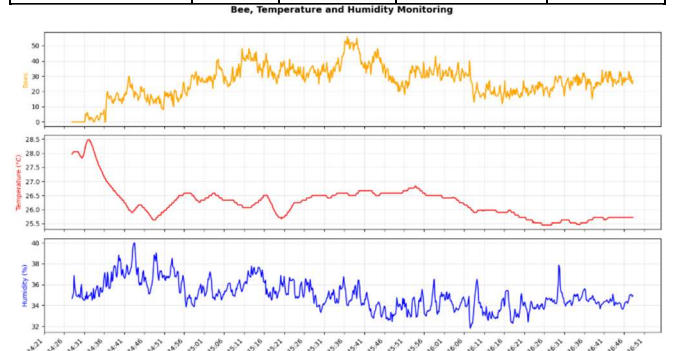


Fig. 10. Collected data

A gold-standard validation study was conducted to systematically compare bee counts generated by the proposed model against manual reference counts. Pearson correlation analysis (Figure 11) was performed using a sample of 50

images to assess the association between the automated and manual counting methods. The correlation coefficient was $r = 0.93$ ($p < 0.00001$), indicating a strong positive correlation with high statistical significance. To further evaluate the agreement between automated and manual counts, a Bland-Altman [12] analysis was performed using the same 50 images (Figure 12). The study revealed a small mean bias of 3.4 bees between the two methods, with 95% limits of agreement of ± 1.96 SD. The results demonstrated that the majority of differences between the automated and manual approaches fell within the established boundaries of agreement, confirming the reliability and robustness of the proposed model as a practical tool for automated bee monitoring applications

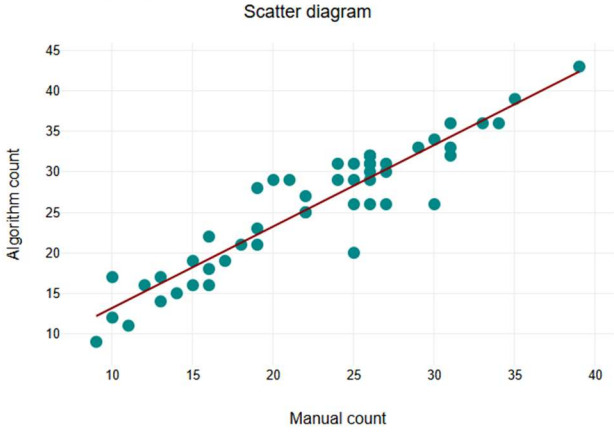


Fig. 11. Scatter diagram

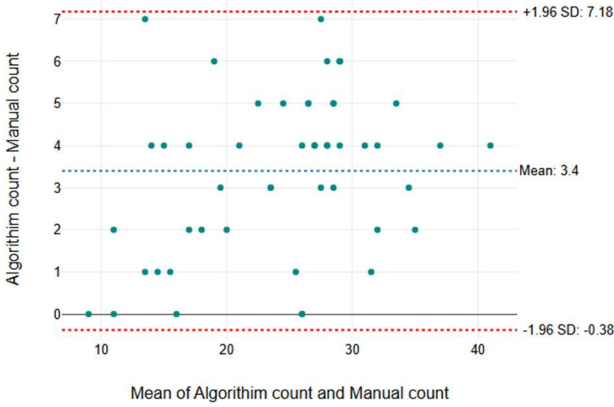


Fig. 12. Bland-Altman plot

V. CONCLUSIONS

A comprehensive end-to-end workflow was implemented for the development of the proposed bee monitoring system. The methodology began with the creation of a dedicated dataset, requiring several months of continuous data collection, preprocessing, and annotation to ensure both representativeness and statistical robustness across diverse environmental conditions. The trained model was implemented using the YOLOv8 architecture and deployed on a Raspberry Pi Zero 2W embedded platform to evaluate real-world performance in resource-constrained conditions. Performance evaluation demonstrated an average inference time of 908 ± 61.5 ms per frame (mean $\pm$ SD) and achieved a

mean accuracy of 93% across the validation dataset. These computational performance metrics demonstrate the practical feasibility of the proposed approach for monitoring applications in field conditions, particularly when operating under the computational and power constraints typical of low-cost embedded systems.

Performance evaluation confirmed the reliability of the system through both correlation analysis and Bland-Altman assessment, which collectively demonstrated strong statistical agreement between automated and manual bee counts. These results provide evidence that the model demonstrated the ability to estimate the number of bees at the hive entrance. This included bees that were stationary, walking, or in flight, as well as bees with partially visible bodies.

This capability is particularly valuable for precision beekeeping applications, where continuous and autonomous monitoring can facilitate early detection of behavioral anomalies, support proactive hive health management strategies, and generate comprehensive datasets for broader ecological research. The automated nature of the system eliminates the need for manual observation, reducing labor costs and human disturbance to colonies while maintaining high measurement accuracy.

Despite these promising results, limitations were identified in the current approach. The system exhibited occasional false positives and false negatives, particularly under challenging conditions including overlapping bee trajectories, rapid movement patterns, shadowing effects, complex background imagery, and variable lighting conditions. Furthermore, while the inference speed remains acceptable for real-time monitoring applications, it may present scalability constraints when deployed across larger apiaries or when operating at higher frame rates.

Several research directions will be pursued to address current limitations and enhance system capabilities. First, an energy consumption optimization model will be developed to minimize power requirements, enabling fully autonomous and continuous operation. This approach will explore the integration of renewable energy sources, particularly photovoltaic systems, to achieve energy self-sufficiency in field deployments. Second, the system architecture will be extended to incorporate a cloud-based communication infrastructure, facilitating remote monitoring capabilities and enabling large-scale data aggregation across distributed apiaries. This enhancement will support advanced analytics and cross-hive comparative studies. Third, systematic optimization of sampling rates will be conducted to identify the optimal balance between inference latency, computational efficiency, and detection accuracy. These improvements are anticipated to significantly enhance system scalability and robustness across diverse environmental conditions, thereby strengthening its potential for precision apiculture applications and broader ecological monitoring initiatives.

VI. REFERENCES

- [1] Khalifa, S.A.M.; Elshafiey, E.H.; Shetaia, A.A.; El-Wahed, A.A.A.; Algethami, A.F.; Musharraf, S.G.; AlAjmi, M.F.; Zhao, C.; Masry, S.H.D.; Abdel-Daim, M.M.; et al. Overview of Bee Pollination and Its Economic Value for Crop Production. *Insects* 2021, *12*, 688.
- [2] Astuti, P.K.; Ayoob, A.; Strausz, P.; Vakayil, B.; Kumar, S.H.; Kusza, S. Climate change and dairy farming sustainability; A causal loop paradox and its mitigation scenario. *Heliyon* 2024, *10*, e25200.
- [3] Tabor, J.A.; Koch, J.B. Ensemble Models Predict Invasive Bee Habitat Suitability Will Expand under Future Climate Scenarios in Hawai'i. *Insects* 2021, *12*, 443.
- [4] Martínez-López, O.; Koch, J.B.; Martínez-Morales, M.A.; Navarrete-Gutiérrez, D.; Enríquez, E.; Vandame, R. Reduction in the potential distribution of bumble bees (Apidae: Bombus) in Mesoamerica under different climate change scenarios: Conservation implications. *Glob. Chang. Biol.* 2021, *27*, 1772–1787.
- [5] Lanner, J.; Dubos, N.; Geslin, B.; Leroy, B.; Hernández-Castellano, C.; Dubai'c, J.B.; Bortolotti, L.; Calafat, J.D.; 'Cetkovi'c, A.; Flaminio, S.; et al. On the road: Anthropogenic factors drive the invasion risk of a wild solitary bee species. *Sci. Total Environ.* 2022, *827*, 154246.
- [6] Redmon, J. You only look once: Unified, real-time object detection. In Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition, Las Vegas, NV, USA, 27–30 June 2016.
- [7] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436-444.
- [8] Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill Education.
- [9] Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- [10] Russell S, Norvig P (2009) *Artificial intelligence: a modern approach*, 3rd ed. Prentice Hall, Upper Saddle River
- [11] <https://roboflow.com/> (accessed on 24 september 2025).
- [12] Martin Bland, J., Altman, D.G., 1986. Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet* 327, 307–310. [https://doi.org/10.1016/S0140-6736\(86\)90837-8](https://doi.org/10.1016/S0140-6736(86)90837-8).